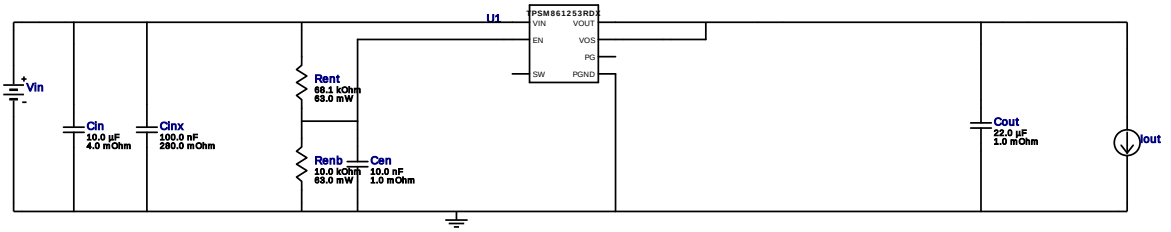
WEBENCH ® Design Report

VinMin = 12.0V
VinMax = 12.0V
Vout = 3.3V
Iout = 1.0A

Device = TPSM861253RDX
Topology = Buck
Created = 2025-01-24 08:21:19.284
BOM Cost = $0.69
BOM Count = 7
Total Pd = 0.22W

Design : 3 TPSM861253RDX
TPSM861253RDX 12V-12V to 3.30V @ 1A



Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cen	MuRata	GRM155R71C103KA01D Series= X7R	Cap= 10.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cin	MuRata	GRM21BR61E106MA73L Series= X5R	Cap= 10.0 uF ESR= 4.0 mOhm VDC= 25.0 V IRMS= 2.8 A	1	$0.04	0805 7 mm²
Cinx	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cout	MuRata	GRM188R60J226MEA0D Series= X5R	Cap= 22.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 6.0 A	1	$0.04	0603 5 mm²
Renb	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rent	Yageo	RC0402FR-0768K1L Series= ?	Res= 68.1 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	TPSM861253RDX	Switcher	1	$0.57	RDX0007A-MFG 22 mm²

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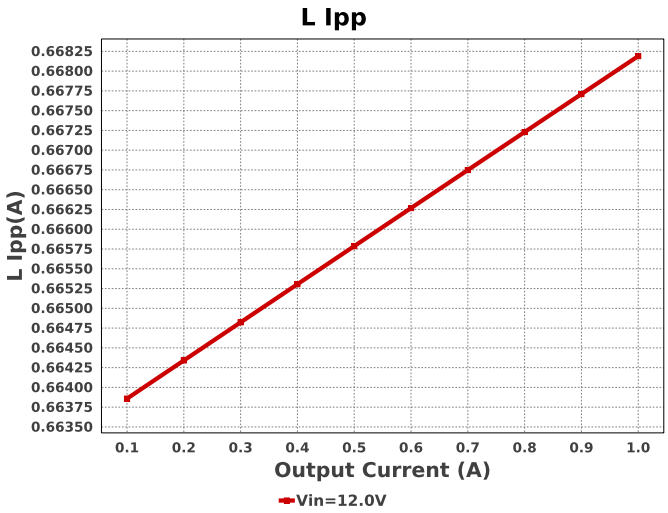

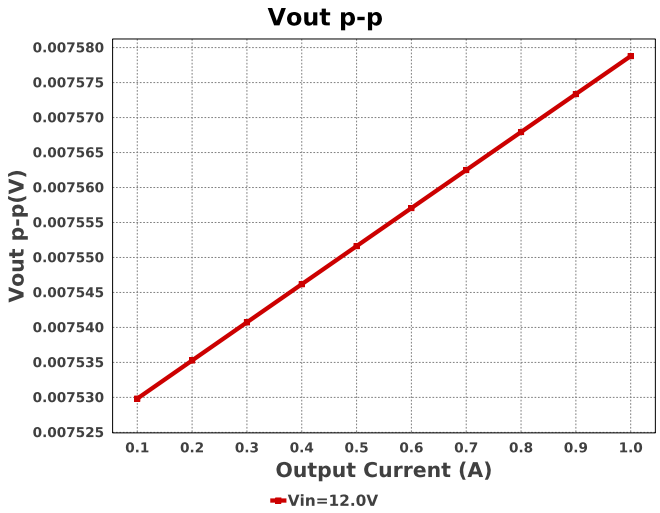

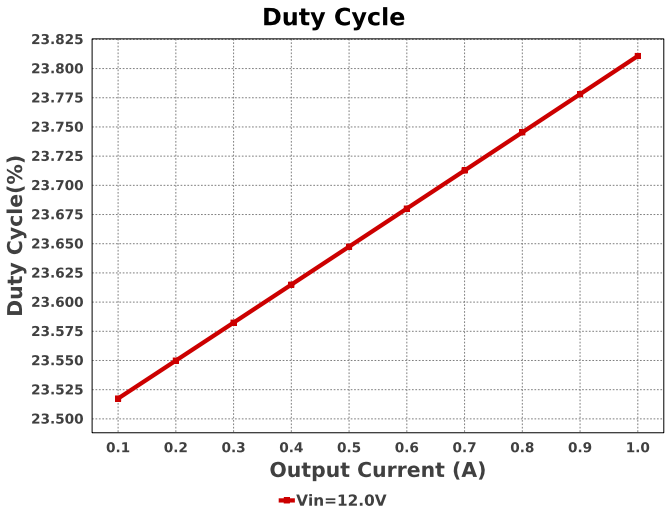

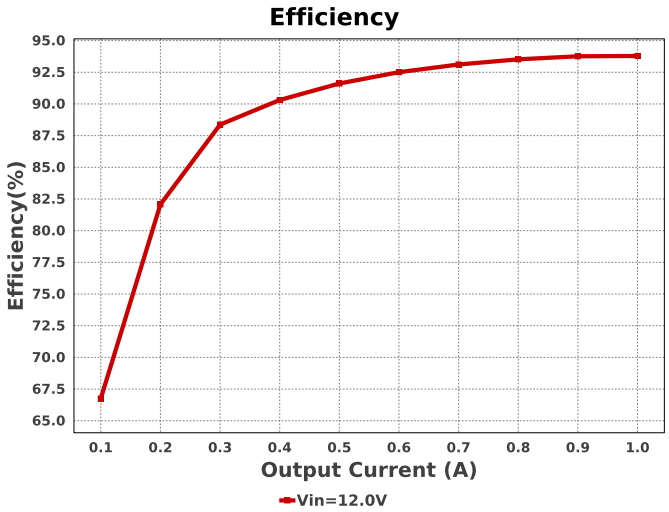

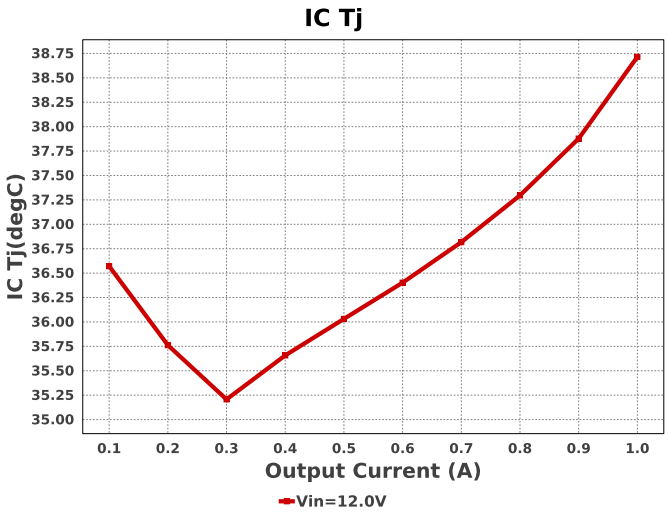

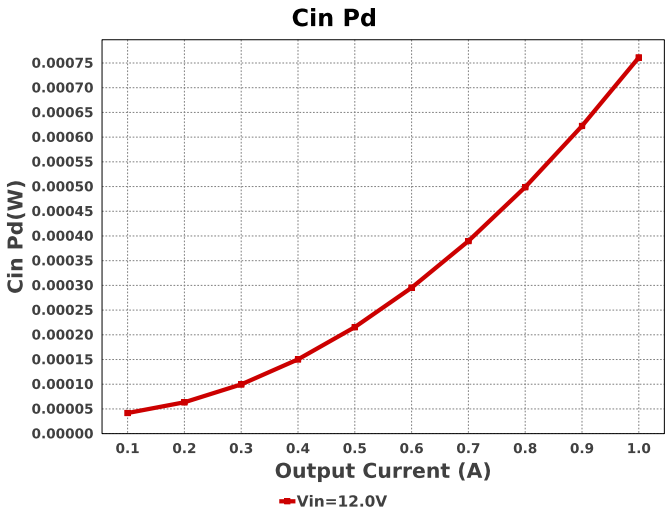

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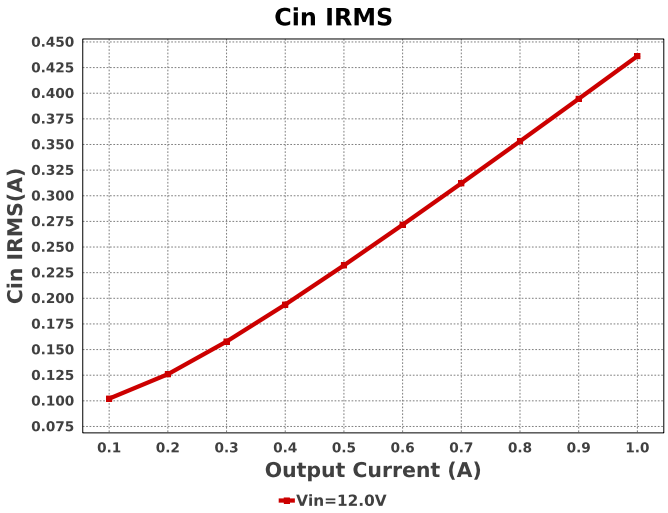
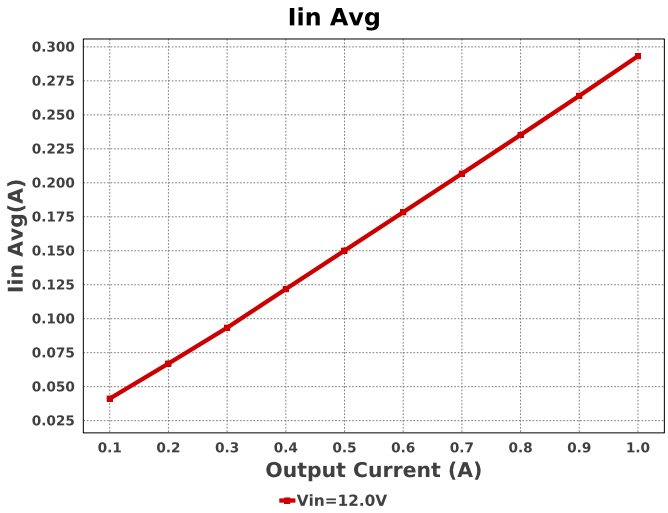
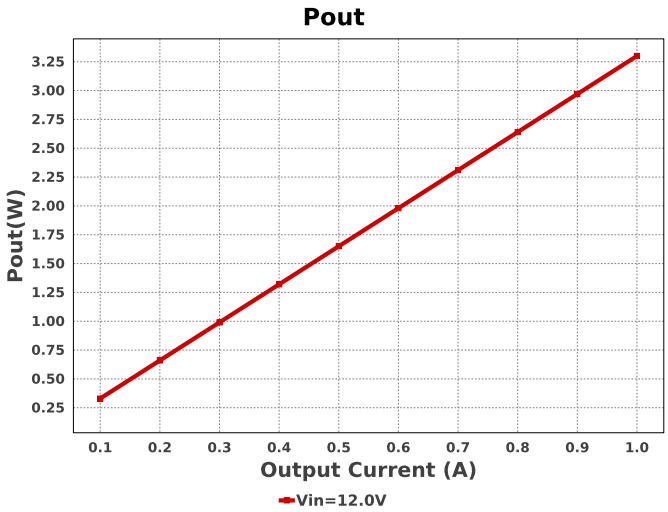
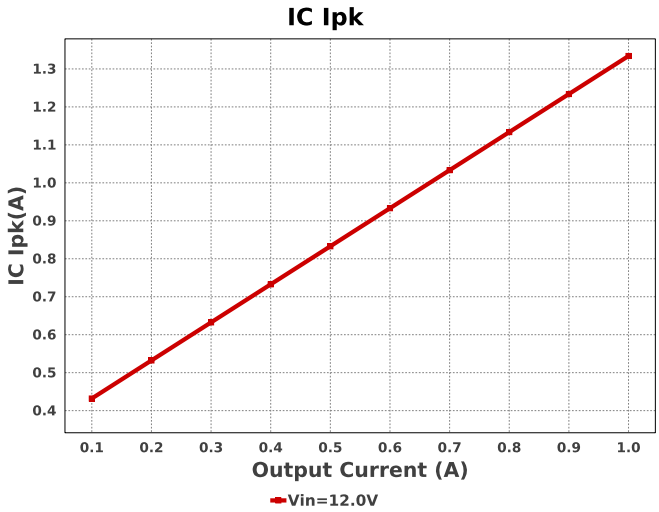
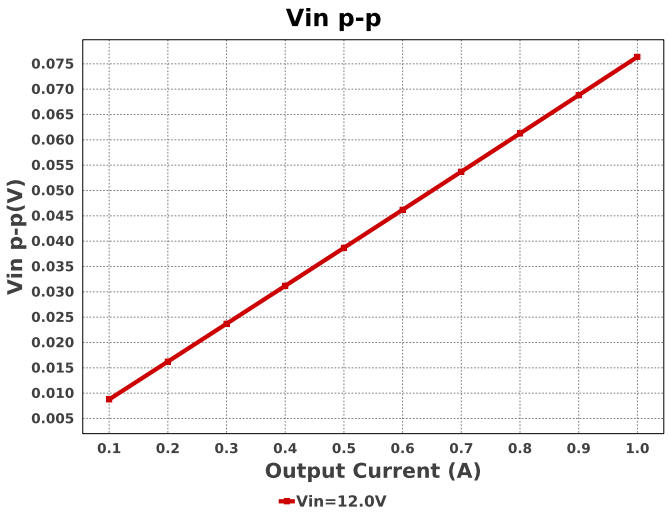
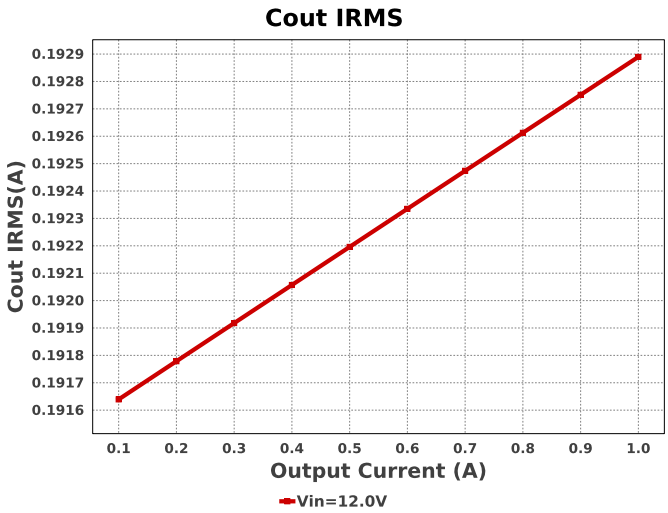

Cin IRMS
Cin IRMS(A)
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=12.0V

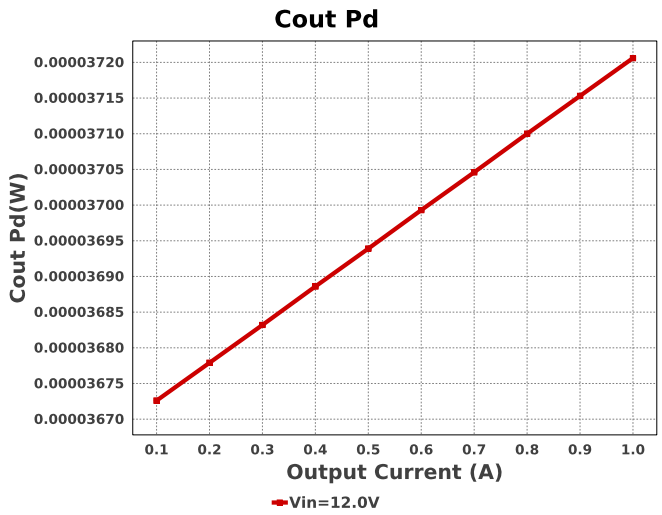

Iin Avg
Iin Avg(A)
0.300
0.275
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=12.0V

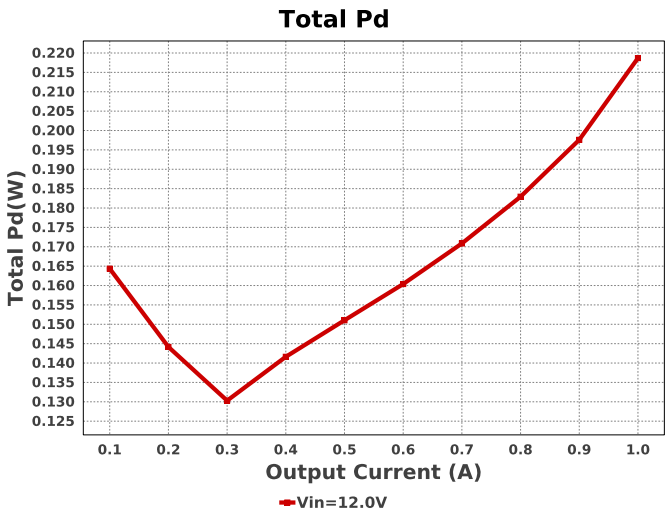

Pout
Pout(W)
3.25
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=12.0V

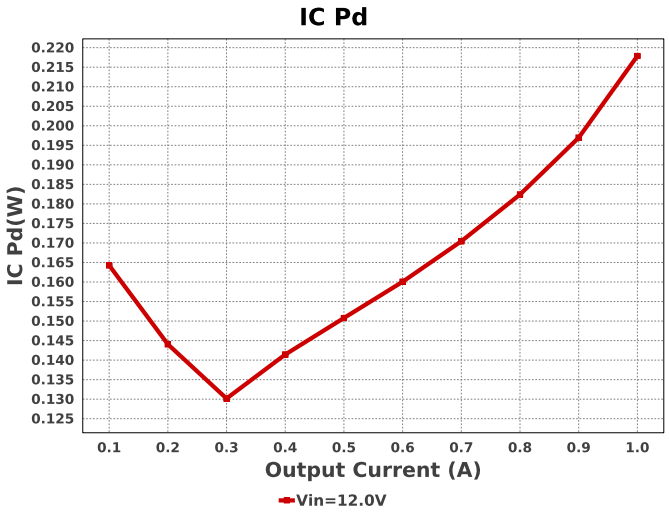

IC Ipk
IC Ipk(A)
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=12.0V


Vin p-p
Vin p-p(V)
0.075
0.070
0.065
0.060
0.055
0.050
0.045
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=12.0V


Cout IRMS
Cout IRMS(A)
0.1929
0.1928
0.1927
0.1926
0.1925
0.1924
0.1923
0.1922
0.1921
0.1920
0.1919
0.1918
0.1917
0.1916
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=12.0V







Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	436.201 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	761.09 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	192.89 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	37.206 µW	Capacitor	Output capacitor power dissipation
5.	IC Ipk	1.334 A	IC	Peak switch current in IC
6.	IC Pd	217.87 mW	IC	IC power dissipation
7.	IC Tj	38.715 degC	IC	IC junction temperature
8.	IC Tolerance	9.0 mV	IC	IC Feedback Tolerance
9.	ICThetaJA Effective	40.0 degC/W	IC	IC junction-to-ambient thermal resistance with TI EVM
10.	Iin Avg	293.22 mA	IC	Average input current
11.	Cin Pd	761.09 µW	Power	Input capacitor power dissipation
12.	Cout Pd	37.206 µW	Power	Output capacitor power dissipation
13.	IC Pd	217.87 mW	Power	IC power dissipation
14.	Total Pd	218.683 mW	Power	Total Power Dissipation
15.	BOM Count	7	System Information	Total Design BOM count
16.	Duty Cycle	23.811 %	System Information	Duty cycle
17.	Efficiency	93.785 %	System Information	Steady state efficiency
18.	FootPrint	49.0 mm^2	System Information	Total Foot Print Area of BOM components
19.	Frequency	1.4 MHz	System Information	Switching frequency
20.	Iout	1.0 A	System Information	Iout operating point
21.	L Ipp	668.19 mA	System Information	Peak-to-peak inductor ripple current
22.	Mode	CCM	System Information	Conduction Mode
23.	Pout	3.3 W	System Information	Total output power

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#	Name	Value	Category	Description
24.	Total BOM	$0.69	System Information	Total BOM Cost
25.	Vin	12.0 V	System Information	Vin operating point
26.	Vin p-p	76.373 mV	System Information	Peak-to-peak input voltage
27.	Vout	3.3 V	System Information	Operational Output Voltage
28.	Vout Tolerance	272.73 m%	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
29.	Vout p-p	7.579 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	1.0	Maximum Output Current
VinMax	12.0	Maximum input voltage
VinMin	12.0	Minimum input voltage
Vout	3.3	Output Voltage
base_pn	TPSM861253	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

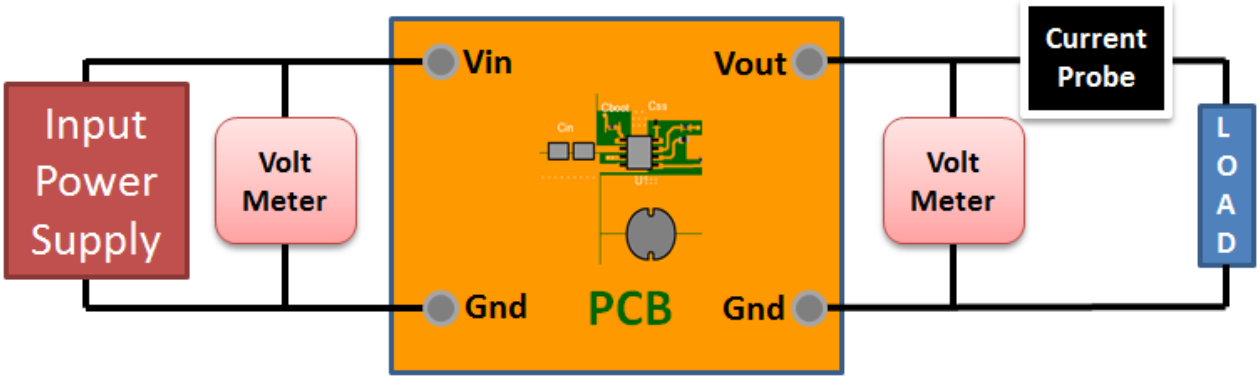
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 12.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.

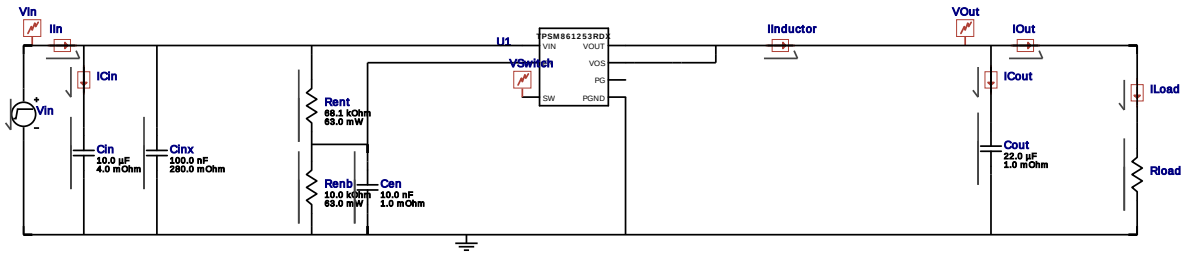


WEBENCH® Electrical Simulation Report

Design Id = 3

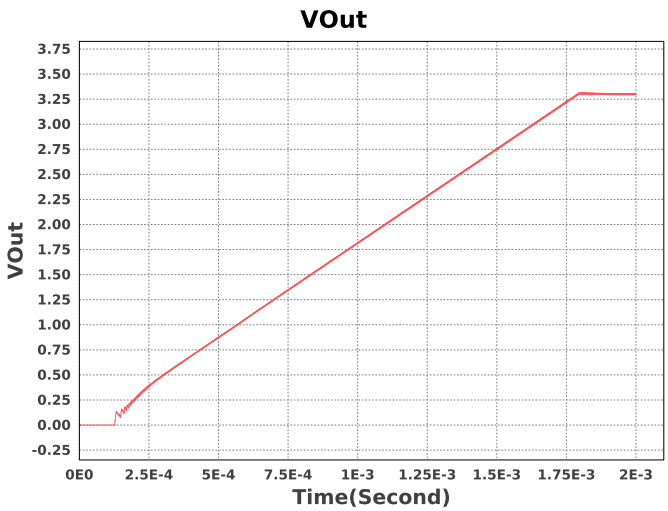
sim_id = 1

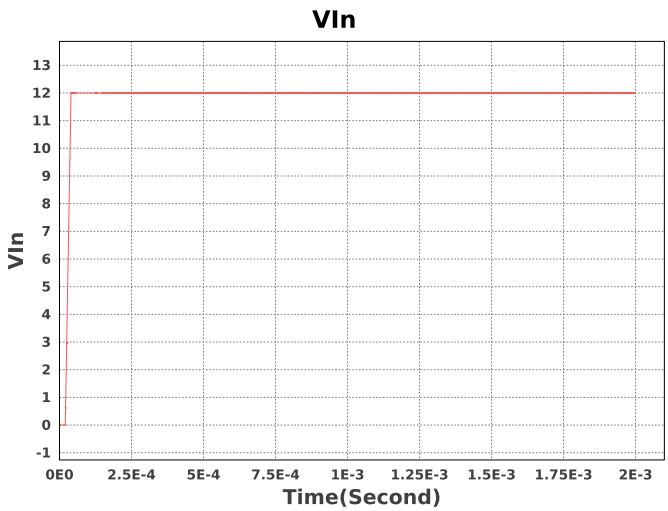
Simulation Type = Startup



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Rload	R	Load Resistance	3.3 ohms





Design Assistance

1. Master key : 164179B73D2D5A7904399F43F8231DB7[v1]

2. **TPSM861253** Product Folder : https://www.ti.com/product/TPSM861253 : contains the data sheet and other resources.

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